09.01.2023

Roll No. ....

TBT–301

B. TECH. (BIOTECHNOLOGY) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

GENETICS AND MOLECULAR BIOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each sub-question carries 10 marks.

1. (a) Elaborate the basis and principles of Mendelian and non-Mendelian genetics. (CO1)

   (b) What is Epigenetics ? Write *three* ways of controlling gene expression and examples of epigenetics. (CO1)

   (c) Write a note on Pedigree analysis and Population genetics. (CO1)

2. (a) Describe how the organization of prokaryote and eukaryote genome differ from each other. (CO2)

   (b) Write notes on the following : (CO2)

   (i) Transposons

   (ii) Nucleic acids

   (c) Who discovered the RNA as genetic material and how ? (CO2)

P. T. O.

3. (a) Describe semi-conservative mode of replication and site directed mutagenesis. (CO3)

   (b) What is replication ? What are the steps of replication in prokaryotes ? Elaborate structure of DNA polymerase III. (CO3)

   (c) Discuss the different DNA repair mechanisms (any *three*). (CO3)

4. (a) Write notes on the following : (CO4)

   (i) Gene structure

   (ii) RNA polymerase

   (b) Describe the process and steps of transcription in eukaryotes. (CO4)

   (c) What is post-transcriptional modification ? How does RNA editing occur ? (CO4)

5. (a) Define Genetic code; Wobble hypothesis; Lac operon. (CO5)

   (b) Describe the mechanism of post-translational modifications. (CO5)

   (c) What is gene regulation ? How is it done in eukaryotes ? (CO5)

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TBT–302

B. TECH. (BIOTECHNOLOGY) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

CELL BIOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the packaging of DNA into Chromosome with the help of a diagram. (CO1)

   (b) Describe chemical structure of DNA in detail and draw the chemical structure. (CO1)

   (c) Explain the structure of Prokaryotic and Eukaryotic cell. Write *ten* differences between Eukaryotic cell and Prokaryotic cell. (CO1)

2. (a) Explain the structure and function of Mitochondria with special emphasis on Electron Transport Chain. (CO1, CO2)

   (b) Describe the process of protein synthesis in detail with appropriate diagram. (CO1, CO2)

   (c) Discuss the fluid mosaic model of Plasma membrane. (CO1, CO2)

P. T. O.

3. (a) Explain the different phases of meiosis with appropriate diagram. (CO3, CO4)

   (b) Explain the mechanism of cell signaling. (CO3, CO4)

   (c) What is cancer ? Explain the molecular basis of cancer. (CO3, CO4)

4. (a) Explain the mechanism of cell death with appropriate diagram. (CO4)

   (b) What is cell cycle ? Explain the types of cell division. (CO4)

   (c) Explain the mechanism of cell cycle regulation. (CO4)

5. (a) Write short notes on the following : (CO5)

      (i) FASTA

      (ii) BLAST

   (b) Explain the working principle and components of compound microscope. (CO5)

   (c) What is FISH technique ? Explain its uses in biotechnology field. (CO5)

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TBT–303

B. TECH. (BIOTECHNOLOGY) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

MICROBIOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Write short notes on the following : (CO1/CO2)

(i) Numerical Taxonomy

(ii) Biogenesis Theory

(b) Describe the mechanism of transfer of plasmid DNA in prokaryotes with suitable examples. (CO1/CO2)

(c) Discuss the different branches of Microbiology. (CO1/CO2)

2. (a) A contaminated milk sample is given to you. Explain the suitable culture method for the isolation of bacteria. (CO3)

(b) Describe serial dilution technique and its application using suitable diagram. (CO3)

(c) Give different types of media based on their functions. (CO3)

P. T. O.

3. (a) Explain the method of measuring microbial growth. (CO4/CO5)

   (b) Explain the role of UV light used in Laminar air flow to maintain sterility. (CO4/CO5)

   (c) Give the mathematical expression of growth and establish relation between G and K. (CO4/CO5)

4. (a) Write short notes on the following : (CO3/CO4)

   (i) Virion

   (ii) Bacteriophage

   (b) Describe the method of virus cultivation in a fertile egg. Give suitable diagram. (CO3/CO4)

   (c) How are the lytic and lysogenic cycle interconnected in phage ? Give suitable diagram. (CO3/CO4)

5. (a) A culture media powder is given to you. It is asked to prepare 200 ml media using 13 gm powder. Give the method of its preparation and sterilization. (CO3/CO5)

   (b) Write short notes on the following : (CO3/CO5)

   (i) EMB media

   (ii) MSA media

   (c) Give any method to identify the bacterial motility in Microbiology lab. Also make suitable diagram. (CO3/CO5)

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TBT–304

B. TECH. (BIOTECHNOLOGY) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

BIOCHEMISTRY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Elaborate on the classification of amino acids based on structure. (CO1)

   (b) What is difference between essential and non-essential amino acids ? Give some examples. (CO1)

   (c) What is oxidative and non-oxidative deamination ? (CO1)

2. (a) Elaborate on the classification of carbohydrates. (CO2)

   (b) Explain TCA cycle and its importance. (CO2)

   (c) Explain the structure of chondroitin sulphate and hyaluronic acid. (CO2)

P. T. O.

3. (a) What is lipid ? Classify. (CO3)

   (b) Give the structure of some fatty acids 15; 3:4, 7, 9 and 18; 2:5, 9. (CO3)

   (c) Explain the beta oxidation of fatty acids. (CO3)

4. (a) Give the classification of enzymes. (CO4)

   (b) Give short notes on holoenzyme, apoenzyme and coenzymes. (CO4)

   (c) Write down the factors affecting enzyme activity. (CO4)

5. (a) What is phosphodiester bond ? Give its structure. (CO5)

   (b) What is de novo biosynthesis of purines and pyrimidines ? (CO5)

   (c) What is purines and pyrimidines ? (CO5)